Roll Number: _____ Name_____ Group_____

THAPAR INSTITUTE OF ENGINEERING &TECHNOLOGY, PATIALA
Department of Computer Science & Engineering
Theory of Computation (UCS701) – Auxiliary Examination
Date: 29/08/2025 & Time: 05:30PM **MM: 100 & MT: 180 Min**

Name of the Faculty: Dr. Chinmaya Panigrahy

Attempt any five of the six questions. Attempt/Answer all sub-parts like (a), (b), (c) for each question in one place. Do mention Page No. of your attempt on the front page. Question paper is of two pages. Please turn over this page to see the questions on the next page.

Ques. No		Questions	Marks	CO	BL
Q1.	(a)	Design a DFA that accepts all the binary strings that (i) Begin with 011. (ii) End with 011. (iii) Contain a substring 011.	12	CO1	L3
	(b)	Write down the regular expressions that represent the languages accepted by the DFAs that you have designed in Q1 (a) (i), (ii), and (iii).	3	CO1	L3
	(c)	List all the strings of length at most four, which are accepted by the DFAs that you have designed in Q1 (a) (i), (ii), and (iii).	5	CO1	L1
Q2.	(a)	What is the difference between a DFA and an NFA? Discuss with the help of the respective tuple representations.	4	CO1	L1
	(b)	Construct an NFA that accepts all the binary strings where the third symbol from the right end is 1.	2	CO1	L3
	(c)	Convert the NFA that you have designed in Q2 (b) to an equivalent DFA.	5	CO1	L4
	(d)	Design a Moore machine to find the occurrence of 1010 in a binary string. Convert the above Moore machine to an equivalent Mealy machine. Show the outputs of the above machines for the input, w = 101010.	9	CO1	L4
Q3.	(a)	Apply Thompson's construction to construct an NFA with epsilon moves for the regular expression, (aa+b)*ab(b+a)*b.	4	CO1	L3
	(b)	Construct a DFA directly from the epsilon NFA obtained in Q3 (a) by applying epsilon-closure method.	8	CO1	L4
	(c)	Simplify the following CFG. $S \rightarrow aA \mid aBB$ $A \rightarrow aAA \mid \varepsilon$ $B \rightarrow bB \mid bbC$ $C \rightarrow B$	5	CO2	L3
	(d)	Prove that the context-free languages are closed under Kleene closure operation.	3	CO2	L3
Q4.	(a)	Consider the language, $L = \{a^n b^n \mid n\}$. For the L, (i) Draw a flow diagram for the corresponding PDA. (ii) Design a PDA to accept L. (iii) What is the tuple representation of the PDA constructed in (ii)?	8	CO2	L3
	(b)	Show that the language, $L = \{0^X \mid X=n^2$, where $n>0\}$ is not context-free using the	5	CO2	L3

		Pumping lemma for context-free languages.			
	(c)	Design a context free grammar to generate all the binary strings with unequal number of 0's and 1's	7	CO2	L4
Q5	(a)	Design a Turing machine for the following language by writing the logic and an equivalent tuple representation. $L=\{a^n b^n c^n \mid n>0\}$	10	CO3	L4
	(b)	Design a post machine for the language given in Q5 (a). Draw the required flow diagram. Show the steps to process the string, w=aabbcc.	10	CO3	L4
Q6	(a)	Discuss various variants of Turing machine available in the literature in detail.	10	CO3	L1
	(b)	Design a Turing machine to multiply two positive integers.	10	CO4	L4

.....All the Best.....